Module 2

Simulink, MATLAB's Control System Functions, and Live Editor

Simulink and Some Control Systems MATLAB Functions Warmup Homework Problems

In this section we define several MATLAB oriented homework problems involving the MATLAB functions, MATLAB live functions, MATLAB live scripts and interactions of either MATLAB scripts or live scripts with Simulink that play important roles in reinforcement learning. The state space models of diverse types of linear dynamic systems are presented mostly in continuous time, but corresponding discrete-time state space models can be obtained by performing discretization using the MATLAB function `c2d`, as explained in Section 2.3 of Module 2. In the cases of linear-quadratic dynamic optimization, the performance criterion weighted matrices are all specified (suggested).

Homework Problem 2.1

For the discrete-time aircraft model considered in Example 2.4 of Module, solve the linear-quadratic optimal control problem assuming that the criterion penalty matrices are identities. First verify that the optimal control exists by checking conditions of Assumption 2.2 stated in Module 2. Solve this problem: (1) writing a MATLAB script, and (2) by using the live functions and live scripts for controllability, observability, and optimality developed in Section 2.4 of Module 2. Assume that all system initial conditions are normalized and equal to one.

Homework Problem 2.2

A continuous-time state space model of a tandem rotor helicopter is given by [1]

$$A = \begin{bmatrix} -0.02 & 0.005 & 2.4 & -32 \\ -0.14 & 0.44 & -1.3 & -30 \\ 0 & 0.018 & -1.6 & 1.2 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0.14 & -0.12 \\ 0.36 & -8.6 \\ 0.35 & 0.009 \\ 0 & 0 \end{bmatrix}$$

Assume that the penalty matrices in the corresponding linear-quadratic optimal control problem are selected as, $R_1 = 2 \times I_4$, $R_2 = 0.3 \times I_2$. Find the eigenvalues of matrix A, and conclude that this system is not open-loop asymptotically stable. Examine the system controllability and observability. Use MATLAB to find the solution of the continuous-time-time algebraic Riccati equations. Find the optimal feedback gain and show that it stabilizes the closed-loop (feedback)

system. Solve this problem by using both the MATLAB scripts and MATLAB live scripts with MATLAB live functions for controllability, observability, and linear-quadratic optimality presented in Section 2.4 of Module 2. Assume that all system initial conditions are normalized and equal to one.

Homework Problem 2.3

Consider a Boeing 747 flight controls system from the paper [2]. Assuming level flight at 40 000 ft at a speed of 774ft/s, and the discretization interval of 1 s , the linearized aircraft dynamic is represented in the discrete-time domain by

$$x(k+1) = Ax(k) + Bu(k)$$

where

$$A = \begin{bmatrix} 0.99 & 0.03 & -0.02 & -0.32 \\ 0.01 & 0.47 & 4.7 & 0.0 \\ 0.02 & -0.06 & 0.4 & 0.0 \\ 0.01 & -0.04 & 0.72 & 0.99 \end{bmatrix}, \qquad B = \begin{bmatrix} 0.01 & 0.99 \\ -3.44 & 1.66 \\ -0.83 & 0.44 \\ -0.47 & 0.25 \end{bmatrix}$$

The penalty matrices in the corresponding linear-quadratic optimal control problem were selected in [2] as, $R_1 = I_4$, $R_2 = I_2$. Use MATLAB and find the solutions of the discrete-time algebraic Lyapunov and Riccati equations. First check whether the unique positive semidefinite solutions of these equations exist by examining the original system asymptotic stability, controllability, and observability of the corresponding pairs. Assume that all system initial conditions are normalized and equal to one.

Homework Problem 2.4

A computer communication network switch can be modeled as a discrete-time linear system with the state space matrices given by [3]

$$A^{(d+2)\times(d+2)} = \begin{bmatrix} 2 & -1 & 0 & \cdots & \cdots & 0 & 1 \\ 1 & 0 & 0 & \cdots & \cdots & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 & -1 \\ \vdots & \vdots & 1 & 0 & \ddots & \vdots & -1 \\ \vdots & \vdots & 0 & 1 & \ddots & 0 & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & 0 & -1 \\ 0 & 0 & 0 & \cdots & 0 & 1 & -1 \end{bmatrix}, \qquad B^{(d+2)\times 1} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ \vdots \\ \vdots \\ 0 \end{bmatrix}$$

with the input feedback signal given by

$$u(k) = -\alpha_0 x_1(k) - \alpha_1 x_2(k) - \beta_0 x_3(k) - \beta_1 x_4(k) - \cdots - \beta_{d-1} x_{d+2}(k) = -Fx(k)$$

$$\sum_{j=0}^{1} \alpha_j > 0, \qquad \sum_{i=0}^{d} \beta_i = 0,$$

$x(k)$ is the buffer queue length, $u(k)$ represents the service rate of a buffer, d is the round trip delay in discrete-time units between the source and the switch, and α_j and β_i are the feedback gains to be chosen by the control (network) engineer. The dynamics of the packet arrival rate can be modelled by the following first-order difference equation $r(k+1) = r(k) + u[k]$.

(a) Assume that $d = 12$, and find the feedback gain using the MATLAB function `place` such that the closed-loop system is asymptotically stable, that is, $|\lambda_i(A - BF)| < 1, \forall i$. In other words, select all 14 closed-loop eigenvalues to lie within the unit circle, defined the corresponding row vector "lambda_desired", and find the corresponding feedback gain using the following MATLAB statement: `>> F=place(A,B,lambda_desired)`. In order to use the MATLAB function `place`, it is necessary to check system controllability via the MATLAB function `ctrb(A,B)`.

(b) Assume that the weighted state and control matrices in the corresponding quadratic performance criterion are identities, that is `>> Q=eye(14)` and `R=1` solve the discrete-time linear-quadratic optimal control problem using the MATLAB function `dlqr`. Find the value for the optimal performance criterion assuming that all initial conditions are equal to 1.

Homework Problem 2.5

The national income model can be represented by the following relations [3]

$$y(k) = c(k) + i(k) + f(k)$$
$$c(k) = \alpha y(k-1)$$
$$i(k) = \beta(c(k) - c(k-1))$$

where $y(k)$ is the national income, $c(k)$ represents consumer expenditures, $i(k)$ is induced private investment, $f(k)$ represents government expenditures, and α, β are positive constants. These three equations lead to the second-order difference equation for the national income

$$y(k+2) - \alpha(1+\beta)y(k+1) + \alpha\beta y(k) = f(k+2)$$

Choose several sets of values for parameters α, β. Build a Simulink block diagram using the "unit delay" MATLAB blocks to simulate dynamic behavior of this system and find its step response. Plot the system response over the period of ten years, assuming that the discrete-time interval is in quarters ($k = 0,1,2...,36$). For simplicity assume zero-initial conditions. Obtain the state space model for the national income model and perform simulation using the MATLAB state space block. Find the impulse and step responses, as well as the response when the system input is a sum of step and ramp signals as $f[k] = \text{step} + \gamma \times \text{ramp}$, with parameter γ taking small values such as $0.01, 0.05, 0.5$.

Homework Problem 2.6

Write the main program and the live function (subprogram) for solving the linear-quadratic optimal control in discrete-time presented in Section 2.3 of Module 2, including controllability and observability tests of the required problem matrices. For controllability and observability live functions, you may use the MATLAB programs from Example 2.1 of Module 2. Apply the developed programs to Boeing 747 aircraft whose state space matrices are presented in the homework Problem 2.3. Evaluate the optimal performance assuming that all system initial conditions are equal to one.

Hint: The corresponding continuous-time problem is solved in Example 2.2 of Module 2.

Homework Problem 2.7

Consider the simple pendulum from Exercise 2.4 of Module 2. Build the pendulum nonlinear model Simulink bock diagram. Use the MATLAB function `linmod` to find the pendulum linearized model state space matrices at the operating (nominal) points equal to the steady state points.

Homework Problem 2.8

A two-link robotic manipulator with a flexible joint is represented by the following nonlinear control system [3]

$$\begin{aligned}
&\frac{dx_1(t)}{dt} = x_2(t)\\
&\frac{dx_2(t)}{dt} = -\frac{mgl}{I}\sin(x_1(t)) - \frac{k}{I}(x_1(t) - x_3(t))\\
&\frac{dx_3(t)}{dt} = x_4(t)\\
&\frac{dx_4(t)}{dt} = \frac{k}{J}(x_1(t) - x_3(t)) + \frac{1}{J}u(t)\\
&y(t) = x_1(t)
\end{aligned}$$

where $x_1(t)$ and $x_3(t)$ are link angular positions and $x_2(t)$ and $x_4(t)$ are the corresponding angular velocities. I and J are moments of inertia, m and l are link mass and length, and k is the link spring constant. Assume that the problem normalized numerical values are given by $mgl = 5, I = J = 1, k = 0.08$ and that the desired value for the steady state output is $y_{ss} = x_{1ss} = \pi / 3$. Find the steady state values for the remaining system state variables and for the system control input. Build the Simulink block diagram for this system and use the MATLAB function `linmod` for find the linearized system state space matrices at the steady state points. Check stability, controllability and observability of the obtained linearized system.

Hint: See Exercise 2.4 of Module 2.

Homework Problem 2.9

An islanded electric microgrid power system is represented by the following continuous-time state space matrices [4]

$$B=\begin{bmatrix}0 & 0 & 3333.3 & 0 & 0 & 0\\ 0 & 0 & 0 & 3333.3 & 0 & 0\end{bmatrix}^T,\quad C=\begin{bmatrix}0 & 1 & 0 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0 & 0\end{bmatrix},\quad D=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$$A=\begin{bmatrix}-209.32 & 376.99 & 15908 & 0 & -15908 & 0\\ -376.99 & -209.32 & 0 & 15908 & 0 & -15908\\ -3333.3 & 0 & -5 & 376.99 & 0 & 0\\ 0 & -3333.3 & -376.99 & -5 & 0 & 0\\ 8.9366 & 0 & 0 & 0 & -3.1416 & 376.9\\ 0 & 8.9366 & 0 & 0 & -376.99 & -3.1416\end{bmatrix}$$

Solve the LQ optimal control problem with the weighted matrices $R_1 = I_6, R_2 = I_2$ and find the optimal feedback gain and the optimal performance criterion value using the MATLAB live functions presented in Examples 2.1 and 2.2 of Module 2. Assume that all system initial conditions are normalized and equal to one.

Homework Problem 2.10

A double fed induction generator (DFIG) wind turbine mathematical model has been studied in the continuous-time state space form [5]. Discretize this system using the sampling rate $T_s = 0.01\,\text{s}$. Examine controllability and observability of the obtained discrete-time system using the MATLAB live functions developed in Section 2.4 of Module 2. Examine the system open-loop stability by finding the eigenvalues of matrix A using `>> eig(Ad)`.

$$A=\begin{bmatrix}-25.17 & 5717 & -89.73 & -5241 & -212.5 & 1617 & 0 & 80.9\\ -5711 & -27.88 & 5242 & -89.73 & 1646 & 0 & 1617 & 0\\ -24.38 & 5222 & -91.82 & -5050 & -217.4 & 1655 & 0 & 82.8\\ -5517 & -27.16 & 5050 & -91.82 & 1726 & 0 & 1655 & 0\\ 0.207 & -0.123 & -0.194 & -0.0198 & 0 & 0 & 0 & 0\\ 0.438 & 0.070 & -10 & 0 & 0 & 0 & 0 & 10\\ 0.635 & 0.102 & 0 & -10 & 9.17 & 0 & 0 & 0\\ -1.21 & 0.194 & 0 & 0 & 0 & 0 & 0 & 0\end{bmatrix}$$

$$C=\begin{bmatrix}0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0\end{bmatrix},\quad D=\begin{bmatrix}0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0\end{bmatrix}$$

$$B = \begin{bmatrix} 1658 & 0 & 1617 & 0 & 0 \\ 0 & -1658 & 0 & 1617 & 0 \\ -1617 & 0 & 1655 & 0 & 0 \\ 0 & -1617 & 0 & 1655 & 0 \\ 0 & 0 & 0 & 0 & 0.1429 \\ -10 & 0 & -67.47 & 70 & 0 \\ 0 & -10 & -8.256 & 0 & 10.24 \\ 0 & 0 & -7 & 7 & 0 \end{bmatrix}$$

Homework Problem 2.11

Consider an F-15 aircraft mathematical model represented in continuous-time in its state space form with the corresponding quadratic performance criterion weighted matrices defined in [6] as

$$A = \begin{bmatrix} -0.00819 & -25.70839 & 0 & -32.17095 \\ -0.00019 & -1.27626 & 1 & 0 \\ 0.00069 & 1.027626 & -2.40523 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \; B = \begin{bmatrix} -6.80939 \\ -0.14968 \\ -14.06111 \\ 0 \end{bmatrix}, \; Q = I_4, \; R = 1$$

Check whether controllability-observability Assumption 2.1 of Module 2 is satisfied, and if yes, use MATLAB find the optimal feedback gain and the optimal value of the quadratic performance criterion. Assume that all system initial conditions are normalized and equal to one.

Homework Problem 2.12

A linearized state space model of a *communication satellite* is defined by the following continuous-time state space matrices [7]-[8]

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3\omega^2 & 0 & 0 & 0 & 2\omega & 0 \\ 0 & 0 & 0 & -2\omega & 0 & 0 \\ 0 & 0 & -\omega^2 & 0 & 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0_{3\times 3} \\ I_{3\times 3} \end{bmatrix}, \quad C = \begin{bmatrix} I_{3\times 3} & 0_{3\times 3} \end{bmatrix}, \quad D = 0_{3\times 3}$$

where ω represents the orbital angular velocity. The first three state space variables are positions respectively in $x(t), y(t), z(t)$ directions, and the remaining three coordinates are the corresponding velocities $\dot{x}(t), \dot{y}(t), \dot{z}(t)$. Note that the same state space model can be used also

for a *spacecraft* orbiting the earth [7]. The orbital angular velocity is $\omega = 7.222 \times 10^{-5}$ rad/s and the initial conditions are $x(0) = \begin{bmatrix} 10000 & 10000 & -10000 & 2 & 1 & 3 \end{bmatrix}^T$, as taken in [8]. Examine controllability and observability of this system for a realistic range of orbital angular velocity using the developed live functions in Section 2.4.1, Example 2.1 of Moule 2. Write a MATLAB live script with a slider that cover the range of used orbital angular velocities. Discretize this system using MATLAB function `c2d`. Solve the corresponding discrete-time linear-quadratic optimal control problem assuming identity penalty matrices.

Homework Problem 2.13

Solve continuous-time homework Problem 2.11 using MATLAB's interplay between live scripts and Simulink by following the corresponding methodology as presented in Section 2.4.3 in Module 2.

Homework Problem 2.14

Solve discrete-time Problem 2.3 using MATLAB's interplay between live scripts and Simulink by following the corresponding methodology as presented in Example 2.4 in Module 2.

Homework Problem 2.15

A continuous-time LQ controller for a linearized model of a robotic manipulator called the acrobot was considered in [9] assuming unity penalty matrices. The fourth dimensional system with one control input is defined by the following state space matrices

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 12.49 & -12.54 & 0 & 0 \\ -14.49 & 29.36 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ -2.98 \\ 5.98 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix}, \quad D = 0$$

Discretize this system using the sampling rate $T_s = 0.01$ and solve the corresponding linear-quadratic discrete-time problem assuming the penalty matrices remain the same. Assume that all system initial conditions are normalized and equal to one.

Homework Problem 2.16

A linearized mathematical model of an F-16 aircraft can be found in [10]. It is represented by the state space equations that describe aircraft's lateral dynamics. The state space matrices are given by

$$A = \begin{bmatrix} -0.1203 & 0.1479 & -0.9860 & 0.0636 \\ -17.5184 & -1.2523 & 0.5484 & 0 \\ 3.5318 & -0.0274 & 0.1788 & 0 \\ 0 & 1 & 0.1489 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0.0001 & 0.0003 \\ -0.2560 & -0.0044 \\ -0.0133 & -0.0292 \\ 0 & 0 \end{bmatrix}$$

$$C = I_4, \qquad D = 0_{4\times 2}$$

Take for the penalty matrices $R_1 = 10 \times I_4$, $R_2 = I_2$, test controllability and observability assumptions and if they are satisfied solve the corresponding LQ optimal control problem. Assume $x(0) = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$.

References

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